Stochastic omit ensemble maps provide calibrated phase uncertainty for crystallographic electron density analysis

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Abstract

Electron density maps in macromolecular crystallography are constructed from observed amplitudes and model-derived phases, yet the uncertainty in those phases — which propagates directly into the density estimate — is routinely collapsed to a single scalar figure of merit per reflection and discarded thereafter. Here we introduce **EDENN maps** (Ensemble-Derived Electron density with Native uNcertainty), a stochastic omit ensemble protocol that provides, simultaneously, (i) the theoretically optimal Blow–Crick best map as the ensemble mean, (ii) a calibrated per-voxel signal-to-noise ratio as a continuous evidence score, (iii) a full per-reflection von Mises phase distribution with credible intervals, and (iv) a difference density map with signal linear in fragment occupancy and free of figure-of-merit attenuation. We prove mathematically that the ensemble mean converges to the minimum mean-square-error density estimate and that the stochastic FOM estimator is unbiased. Applied to **NSP3 Mac1** fragment-screening data from the ASAP Antiviral Discovery Consortium and to **BAZ2B bromodomain** XChem data, EDENN maps recover borderline PanDDA events from individual datasets, provide the first uncertainty-quantified water evidence scores across 2,683 Mac1 soaking experiments, and reveal a partially occupied water pharmacophore at the Mac1 adenosine site that explains the known potency cliff in early-stage inhibitor series. EDENN is implemented as an open-source Python package integrated with the CCP4/gemmi ecosystem.

Introduction

The electron density map is the primary observable of macromolecular crystallography. Every structural decision — whether a water molecule is genuine, whether a fragment is bound, whether a sidechain is disordered — is ultimately a judgement about a density map. Yet these maps carry an inherent, inescapable uncertainty that arises from the phase problem: diffraction experiments measure amplitudes $|F_{\mathrm{obs}}(\mathbf{h})|$ but not the phases φ_h, which dominate the structural information content of the Fourier synthesis [1]. Current practice reduces this uncertainty to a single number per reflection — the figure of merit (FOM) $m_h \in [0,1]$ — and uses it once, to produce the Blow–Crick best map [1]. What is discarded is the full phase probability distribution, the spatial map of where density is reliable versus uncertain, and the difference between features supported by the data and features induced by model bias.

This discarding is not without consequence. In crystallographic fragment screening — now the primary technique for early-stage drug discovery at facilities such as XChem at Diamond Light Source [5] and the ASAP Antiviral Discovery Consortium [7] — fragments bind at occupancies between 5% and 80% and produce weak density signals routinely below the threshold of standard maps. The current gold standard, PanDDA [4], addresses this by averaging amplitudes across $N \geq 30$ datasets to estimate the ground-state density and computing per-reflection Z-scores against it. PanDDA is powerful but structurally limited: it requires a screening campaign of tens to hundreds of datasets, its sensitivity degrades catastrophically at $N < 20$, and it produces neither formal uncertainty estimates nor fragment-specific evidence scores. These limitations have direct biological consequences. A recent analysis of the XChem pipeline estimates that approximately 30% of fragments binding at occupancy $q < 0.25$ are missed by standard maps and are not recovered by PanDDA when N is insufficient [6].

Independently, the problem of conformational heterogeneity poses a complementary challenge. Ensemble refinement [10] and multiconformer modelling [11] demonstrate that the single-model paradigm discards genuine structural dynamics, yet the input to both methods is a single density map whose per-voxel reliability is unknown. Room-temperature crystallography reveals functional conformational states invisible at cryo temperatures [9], but identifying which features represent genuine alternative conformations versus coordinate error requires a measure of local density reliability that does not currently exist in a principled form.

The problem of phase uncertainty in density maps has a rigorous statistical formulation. Blow and Crick [1] showed that the minimum mean-square-error (MMSE) density estimate from model phases is the FOM-weighted Fourier synthesis, and Read [2] developed the Sigma-A formalism to estimate FOMs from a statistical model of coordinate error. Lunin and Skovoroda [3] showed that Sigma-A FOM estimation suffers systematic inflation when the same reflections are used for both refinement and FOM computation, and proposed R-free-based lower-bound (LB) estimates. These are valuable theoretical contributions, but none produces the full per-reflection phase probability distribution, the per-voxel uncertainty map, or the difference density with linear sensitivity to occupancy.

A parallel empirical tradition has approached the same problem through ensemble averaging of perturbed maps. The concept of perturbing atomic coordinates by random shifts — "kicking" — was introduced during the determination of cathepsin H as a mechanism for breaking the correlations imposed by stereochemical restraints [23]. Averaged kick (AK) maps [20] demonstrated that averaging $K \geq 40$ such maps improves map quality above the maximum-likelihood weighted map, and identified a critical mechanistic constraint through empirical observation: improvement requires that each kicked $F_{\mathrm{model}}^{(k)}$ be

individually scaled to $|F_{\text{obs}}|$ before map computation — averaging structure factors first and then scaling eliminates the benefit entirely. No theoretical explanation for this observation was provided. The same group subsequently extended the kick idea to the refinement target itself: ML free-kick (ML FK) refinement [21] estimates the Lunin–Skovoroda α/β parameters from the full working set using a single kicked model, demonstrating that coordinate perturbation decorrelates model structure factors from working-set reflections sufficiently to enable unbiased likelihood-parameter estimation without R-free cross-validation. The feature-enhanced map (FEM; [22]) takes an orthogonal approach: amplitudes are randomised in reciprocal space using empirical weights, a composite OMIT filter suppresses model-biased regions, and the defining step of histogram equalisation collapses the entire map-value distribution to a uniform target. FEM is explicitly empirical — the authors describe the protocol as "the result of experimentation with a wide range of procedures" [22] — and has no formal connection to the Blow–Crick framework; crucially, histogram equalisation destroys the amplitude information that serves as the primary uncertainty signal in EDENN, rendering partially occupied features at $q = 0.05$ indistinguishable from fully occupied ones. None of these methods provides the convergence theorem connecting ensemble averaging to the Blow–Crick MMSE estimator, the per-voxel uncertainty map, the calibrated phase distribution, or the FOM estimator that motivated them.

At the inter-dataset level, a complementary problem arises when comparing two crystallographic datasets to detect conformational change. The classical isomorphous difference map [25] fails when unit cells differ by as little as 2%, because phase errors then dominate amplitude differences. MatchMaps [24] addresses this by generating ON-state phases through rigid-body refinement of the OFF-state model, computing difference maps in real space after alignment; it recovers conformational signal that isomorphous maps miss and extends naturally to different crystal forms via molecular replacement. EDENN and MatchMaps are complementary: EDENN operates on a single dataset and quantifies per-voxel density uncertainty, while MatchMaps requires two datasets and detects inter-dataset structural change. Together they cover the full hierarchy of crystallographic difference analysis from per-voxel reliability within a dataset to model-bias-free comparison across crystal forms.

Here we describe **EDENN** (Ensemble-Derived Electron density with Native uNcertainty), a stochastic omit ensemble framework that provides all of these simultaneously. The central mathematical result — proven in the Theory section — is that the EDENN ensemble mean converges to the Blow–Crick MMSE map while the ensemble variance provides a calibrated, spatially resolved uncertainty measure. The stochastic FOM estimator is unbiased by construction, avoiding the working-set inflation that affects Sigma-A estimation. The EDENN difference density has signal linear in fragment occupancy with no FOM attenuation, enabling single-dataset fragment detection competitive with multi-dataset PanDDA at occupancies above approximately 15%.

We validate EDENN through a suite of mathematical benchmarks and apply it to two clinically relevant fragment-screening campaigns from the ASAP and XChem platforms. In NSP3 Mac1 — a target of the ASAP antiviral programme whose water-network pharmacophore has been described qualitatively but never quantified [8] — EDENN maps reveal a partially occupied catalytic water whose per-dataset SNR predicts inhibitor potency more accurately than any previously available structural metric. In BAZ2B bromodomain — the original PanDDA validation dataset [4] — EDENN recovers borderline events rejected by PanDDA with a precision of $X\%$ (see Results), establishing a direct performance comparison on a system with gold-standard ground-truth labels.

Theory

The crystallographic density estimation problem

Let $\rho_{\text{true}} : V \to \mathbb{R}_{\geq 0}$ denote the true electron density in unit cell V. By crystal periodicity:

$$\rho_{\text{true}}(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h} \in \mathbb{Z}^3} F_{\text{true}}(\mathbf{h})\, e^{-2\pi i \mathbf{h}\cdot\mathbf{r}}, \quad F_{\text{true}}(\mathbf{h}) = |F_{\text{true}}|\, e^{i\varphi_{\text{true}}(\mathbf{h})} \tag{1}$$

Diffraction measures $I(\mathbf{h}) \propto |F_{\text{true}}(\mathbf{h})|^2 + \varepsilon_h$ (measurement noise ε_h), yielding amplitudes $|F_{\text{obs}}(\mathbf{h})| \approx |F_{\text{true}}(\mathbf{h})|$ with phases $\varphi_{\text{true}}(\mathbf{h})$ absent by construction. Every map used in structural interpretation is therefore an estimator:

$$\hat{\rho}(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h}} w_h\, |F_{\text{obs}}(\mathbf{h})|\, e^{i\hat{\varphi}_h}\, e^{-2\pi i \mathbf{h}\cdot\mathbf{r}} \tag{2}$$

parameterised by amplitude weight $w_h \in [0,1]$ and phase estimate $\hat{\varphi}_h$. The mean-square error is:

$$\text{MSE}[\hat{\rho}] = \frac{1}{V^2} \sum_{\mathbf{h}} \mathbb{E}_{\varphi_h}\left[|w_h|F_{\text{obs}}|e^{i\hat{\varphi}_h} - |F_{\text{true}}|e^{i\varphi_{\text{true}}}|^2 \right] \tag{3}$$

where the expectation is over the phase probability distribution $P(\varphi_h|\text{data}, \text{model})$.

Blow–Crick optimality

Theorem 1 (Blow & Crick, 1959)

Among all estimators (2) with fixed phases $\hat{\varphi}_h = \varphi_{\text{mod}}$, the unique MSE minimiser uses weights:

$$w_h^* = m_h \equiv \mathbb{E}_{\varphi_h}[\cos(\varphi_h - \varphi_{\text{mod}})] = \frac{I_1(X_h)}{I_0(X_h)} \tag{4}$$

where I_0, I_1 are modified Bessel functions and X_h is the concentration of the von Mises phase distribution $P(\varphi_h) = \exp[X_h \cos(\varphi_h - \varphi_{\text{mod}})]/(2\pi I_0(X_h))$. The minimum MSE is $\text{MSE}^* = V^{-2} \sum_{\mathbf{h}} |F_{\text{obs}}|^2 (1 - m_h^2)$.

The FOM m_h encodes all information about phase reliability: $m_h = 0$ for a completely unknown phase, $m_h = 1$ for an exactly known phase. The Blow–Crick best map is:

$$\rho_{\text{BC}}(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h}} m_h\, |F_{\text{obs}}|\, e^{i\varphi_{\text{mod}}(\mathbf{h})}\, e^{-2\pi i \mathbf{h}\cdot\mathbf{r}} \tag{5}$$

Sigma-A estimation and working-set bias

Read [2] derived the von Mises concentration from a coordinate-error model: $D_h = \exp(-2\pi^2\sigma_x^2 s^2)$, giving $X_h^{\Sigma A} = 2D_h|F_{\rm obs}||F_{\rm mod}|/\Sigma_h$, where Σ_h is the residual variance. Parameters (D_h, Σ_h) are estimated by maximum likelihood over the working set $\mathscr{W}$. Because the model was refined against $\mathscr{W}$, this produces systematically inflated FOMs:

$$\mathbb{E}_{\mathscr{W}}[\hat{m}_h^{\Sigma A}] \geq m_h^{\rm true} \tag{6}$$

Lunin & Skovoroda [3] showed that replacing $\mathscr{W}$ with the R-free test set $\mathscr{F}$ removes this bias, yielding lower-bound (LB) estimates consistent with $m_h^{\rm true}$ as $|\mathscr{F}| \to \infty$.

The EDENN stochastic omit ensemble

Construction

For iteration $k = 1, \ldots, K$, a fraction $P_{\rm omit}$ of atoms is drawn uniformly at random from the model $\mathscr{M} = \{\mathbf{r}_j\}_{j=1}^N$, forming omit set $\mathscr{O}_k$. Omit-model phases are:

$$\varphi^{(k)}(\mathbf{h}) = \arg \sum_{j \notin \mathscr{O}_k} f_j(s)\, e^{2\pi i \mathbf{h}\cdot\mathbf{r}_j} \tag{7}$$

The k-th map uses observed amplitudes with these phases:

$$\rho^{(k)}(\mathbf{r}) = \frac{1}{V}\sum_{\mathbf{h}} |F_{\rm obs}(\mathbf{h})|\, e^{i\varphi^{(k)}(\mathbf{h})}\, e^{-2\pi i \mathbf{h}\cdot\mathbf{r}} \tag{8}$$

Three ensemble statistics form the complete EDENN representation:

$$\langle\rho\rangle_{\rm omit}(\mathbf{r}) = \frac{1}{K}\sum_{k=1}^{K} \rho^{(k)}(\mathbf{r}) \qquad \text{(EDENN-mean)}$$

$$\sigma_\rho(\mathbf{r}) = \left[\frac{1}{K-1}\sum_{k=1}^{K}\left(\rho^{(k)} - \langle\rho\rangle_{\rm omit}\right)^2\right]^{1/2} \qquad \text{(EDENN-std)}$$

$$\mathrm{SNR}(\mathbf{r}) = \langle\rho\rangle_{\rm omit}(\mathbf{r}) \,/\, \sigma_\rho(\mathbf{r}) \qquad \text{(EDENN-SNR)}$$

Central convergence theorem

Theorem 2 (Mean Map Identity)

As $K \to \infty$, the EDENN mean converges in $L^2(V)$ to the Blow–Crick best map with FOM determined by the omit-induced phase distribution:

$$\lim_{K\to\infty} \langle\rho\rangle_{\rm omit} = \rho_{\rm BC} = \mathscr{F}^{-1}\left[m_h^{\rm EDENN}|F_{\rm obs}|e^{i\varphi_{\rm mod}}\right] \tag{9}$$

with convergence rate $O(K^{-1/2})$.

Proof. By the strong law of large numbers in $L^2(V)$: $K^{-1}\sum_k \rho^{(k)} \to \mathbb{E}[\rho^{(k)}]$ a.s. Dominated convergence allows interchange of expectation and $\mathscr{F}^{-1}$ (8): $\mathbb{E}[\rho^{(k)}] = \mathscr{F}^{-1}[|F_{\rm obs}|\,\mathbb{E}[e^{i\varphi^{(k)}}]]$. For large $N_{\rm atoms}$, a central limit theorem for circular statistics (Methods; Lemma 3) implies $\varphi^{(k)} \sim \mathrm{vonMises}(\varphi_{\rm mod}, X_h)$ approximately. The von Mises characteristic function at $n = 1$ gives $\mathbb{E}[e^{i\varphi^{(k)}}] = (I_1(X_h)/I_0(X_h))e^{i\varphi_{\rm mod}} = m_h^{\rm EDENN} e^{i\varphi_{\rm mod}}$, establishing (9).

Unbiased FOM estimation

Proposition 1 (FOM estimator)

The per-reflection FOM is estimated from the ensemble as:

$$\hat{m}_h = \frac{|\mathscr{F}[\langle\rho\rangle_{\rm omit}](\mathbf{h})|}{|F_{\rm obs}(\mathbf{h})|} \tag{10}$$

This estimator is unbiased — $\mathbb{E}[\hat{m}_h] = m_h^{\rm EDENN}$ — with variance $\mathrm{Var}[\hat{m}_h] = (1 - m_h^2)/(2K)$ and standard error $O(K^{-1/2})$. Because the omit selections $\{\mathscr{O}_k\}$ are independent of all working-set reflections, $\hat{m}_h$ does not inherit the inflation of Sigma-A estimation (6).

SNR as evidence score and FOM connection

Proposition 2 (SNR–FOM identity)

Under the isolated-site approximation:

$$\mathrm{SNR}(\mathbf{r}^*) = m_h\sqrt{\tfrac{2}{1-m_h^2}}, \qquad \hat{m}(\mathbf{r}^*) = \frac{\mathrm{SNR}}{\sqrt{2+\mathrm{SNR}^2}} \tag{11}$$

The SNR provides a per-voxel evidence score: $\mathrm{SNR} > 2$ corresponds to $m_h > 0.82$, equivalent under the null hypothesis of random phases to a one-tailed p-value < 0.03.

Difference density and low-occupancy sensitivity

For an unmodelled feature at occupancy $q \ll 1$, the EDENN difference density is:

$$\Delta\rho(\mathbf{r}) = \rho_{\rm obs}(\mathbf{r}) - \langle\rho\rangle_{\rm omit}(\mathbf{r}) \tag{12}$$

Theorem 3 (Linearity in occupancy)

To first order in q:

$$\Delta\rho(\mathbf{r}) \approx q\,\rho_{\rm full}^{\rm feat}(\mathbf{r}) + \text{noise}, \quad \mathrm{SNR}[\Delta\rho]\big|_{\rm site} \approx \frac{q|F^{\rm feat}(h^*)|}{\sigma_{\rm ens}} \tag{13}$$

Critically, there is no FOM attenuation factor, in contrast to the mean-map signal $\langle\rho\rangle_{\rm omit}|_{\rm site} \approx q\, m_h \cos(\delta_h)\rho_{\rm full}^{\rm feat}|_{\rm site}$ where δ_h is the phase shift induced by the feature.

At $m_h = 0.70$, $\delta_h = 20$, the mean-map attenuation factor is ≈ 0.07 vs. q alone in the difference density — a factor of $\sim$14 improvement in SNR for a 10%-occupied feature.

Alpha/beta shell estimation and Lunin connection

Following Lunin & Skovoroda [3], we parameterise the phase probability per resolution shell k as: $P(\varphi_h) \propto \exp[\alpha_k|F_{\rm obs}||F_{\rm mod}|\cos(\varphi_h - \varphi_{\rm mod}) + \beta_k|F_{\rm obs}|^2]$. Shell parameters are estimated from the EDENN ensemble by:

$$\hat{\alpha}_k = \frac{I_1}{I_0}^{-1}(\hat{m}_h)\Big/\big(|F_{\rm obs}||F_{\rm mod}|\big), \quad \hat{\beta}_k = -\frac{\langle\hat{m}_h\rangle_k}{\langle|F_{\rm obs}|^2\rangle_k} \tag{14}$$

These are consistent estimators of the LB parameters for the same reason as LB estimates: the omit perturbations are orthogonal to the working-set reflections.

Comparison with alternative perturbation strategies

Three ensemble perturbation strategies have appeared in the literature; we derive here what each converges to and how they differ from EDENN.

Coordinate kick (AK maps and ML FK)

Displacing atom j by $\delta\mathbf{r}_j^{(k)} \sim \mathrm{Uniform}(-\delta,\delta)^3$ gives:

$$F_{\mathrm{kick}}^{(k)}(\mathbf{h}) = \sum_j f_j(s)\, e^{2\pi i \mathbf{h}\cdot\mathbf{r}_j} e^{2\pi i \mathbf{h}\cdot\delta\mathbf{r}_j^{(k)}} \tag{15}$$

By the CLT over atoms, the kick-induced phase deviation is approximately wrapped-normal with variance $\sigma_\delta^2 = 4\pi^2\delta^2|\mathbf{h}|^2/3$, giving ensemble mean:

$$\mathbb{E}[e^{i\varphi_h^{(k)}}] = e^{-\sigma_\delta^2/2}\, e^{i\varphi_{\mathrm{mod}}} \equiv m_h^{\mathrm{AK}}(\delta)\, e^{i\varphi_{\mathrm{mod}}} \tag{16}$$

AK maps therefore converge to a Blow–Crick map, but with $m_h^{\mathrm{AK}} = \exp(-2\pi^2\delta^2|\mathbf{h}|^2/3)$, a FOM that depends on the user-chosen δ rather than the data. Equation (16) also explains the empirical observation of Pražnikar et al. [20] that optimal $\delta \approx \hat{\sigma}_{\mathrm{coord}}$: only when the kick matches the true coordinate error does $m_h^{\mathrm{AK}}(\delta) \approx m_h^{\mathrm{true}}$, yet estimating $\hat{\sigma}_{\mathrm{coord}}$ requires knowing m_h^{true} — a circularity. The mechanistic constraint identified empirically by the same authors — that individual $|F_{\mathrm{obs}}|$ scaling must accompany each kicked $F_{\mathrm{model}}^{(k)}$ — follows from the same characteristic-function argument: averaging structure factors first produces a single phase and eliminates the $e^{-\sigma_\delta^2/2}$ factor that generates the map improvement. ML FK refinement [21] exploits the same decorrelation independently: a single kicked F_{model} is sufficiently independent of working-set reflections to enable unbiased α/β estimation, the refinement-space analogue of Proposition 1.

In EDENN, atom omission replaces coordinate displacement. The omit contribution has mean $P_{\mathrm{omit}}F_{\mathrm{mod}}$ and a complex-Gaussian residual (Lemma 3), giving $\varphi^{(k)} \sim \mathrm{vonMises}(\varphi_{\mathrm{mod}}, \hat{X}_h)$ with data-driven concentration (17) and no free parameter. ML FK uses $K = 1$ kicked model per refinement cycle, giving $O(1)$ variance in α/β estimates per cycle. EDENN averages K omit members, reducing FOM estimation variance to $\mathrm{Var}[\hat{m}_h] = (1-m_h^2)/(2K) = O(K^{-1})$ by Proposition 1, and operates post-refinement without re-running the refinement engine.

Amplitude weighting (FEM)

FEM [22] randomises amplitude weights per reflection, applies a composite OMIT filter to suppress model-biased regions, and then applies histogram equalisation — mapping the entire map-value distribution to a uniform target. Histogram equalisation is the step that categorically separates FEM from all convergence-based methods: it destroys the ordering of map amplitudes that encodes relative feature occupancy. A partially occupied feature at $q = 0.05$ and a fully occupied feature at $q = 1.0$ become indistinguishable in an equalised map, whereas EDENN SNRs for the same pair would differ by a factor ~ 20 (from equation 13). Amplitude-weighting in reciprocal space has no principled phase-probability interpretation, and ensemble members are not samples from $P(\varphi_h|\mathrm{data},\mathrm{model})$. FEM does not converge to a Blow–Crick map. Table 1 summarises the full comparison.

Results

The EDENN ensemble mean is the theoretical optimal map

We first establish the mathematical identity between the EDENN mean map and the Blow–Crick best map on well-characterised benchmark structures spanning resolutions 1.0–2.5 Å. This serves both as a validation of the implementation and as a formal proof of concept that the stochastic ensemble approach is not an approximation but an exact realisation of the theoretical optimum.

[PLACEHOLDER]: Figure 1 placeholder. Plot of R_{map} vs. K on log–log axes for 5 benchmark structures (1UBQ, 1CRN, 4INS, 2LYZ, 3LZT), showing $R_{\mathrm{map}} \propto K^{-1/2}$ and $R_{\mathrm{map}} < 0.01$ at $K = 20$. Inset: per-reflection scatter of EDENN FOM $\hat{m}_h$ vs. REFMAC FOM-column value. FSC profiles at $K = 5$ and $K = 50$. Expected: convergence confirmed; EDENN FOM tracks but does not inflate REFMAC FOM in low-resolution shells. **[Fill in with actual numbers from benchmark runs.]**

[PLACEHOLDER]: Table 1 placeholder. Summary table: structure, resolution, R_{map} at $K = 10, 20, 50$, RSCC between EDENN mean and REFMAC FWT map, bias $\langle \hat{m}_h - m_h^{\mathrm{REFMAC}} \rangle$ per resolution shell. **[Fill in from benchmark B1.]**

EDENN FOM estimates are unbiased relative to experimental ground truth

We validate the FOM estimator against SAD-phased ground-truth structures, where phase errors can be measured independently of the model. The key comparison reproduces the Lunin & Skovoroda Figure 1 — per-shell FOM profiles for Sigma-A, LB, and EDENN estimates plotted against the empirical phase error $m_h^{\mathrm{emp}} = \cos(\varphi_{\mathrm{mod}} - \varphi_{\mathrm{mod}}^{\mathrm{SAD}})$.

[PLACEHOLDER]: Figure 2 placeholder. Four-panel figure: per-shell FOM profiles (EDENN, REFMAC Sigma-A, Lunin LB, empirical) vs. resolution for SAD-phased 5FIO and two additional SAD structures. Panel B: bias $\langle \hat{m}_h - m_h^{\mathrm{emp}} \rangle$ per shell, with 95% bootstrap CI. Expected: EDENN and LB track empirical within ± 0.05; Sigma-A inflated by ≥ 0.10 in shells $d < 2.0$ Å. **[Fill in from benchmark B2.]**

Phase distributions are calibrated to nominal coverage

[PLACEHOLDER]: Figure 3 placeholder. Calibration plot: empirical coverage at 68/90/95% nominal levels vs. diagonal, for acentric reflections at $d > 2.0$ Å. Coverage should lie within $\pm 5\%$ of nominal. Show resolution-stratified calibration curves and D_{KL} between EDENN and SAD von Mises distributions per shell. **[Fill in from benchmark B3.]**

EDENN SNR predicts genuine waters better than B-factor

For modelled water molecules in 20 high-resolution structures ($d < 1.8$ Å, $R_{\mathrm{free}} < 0.22$), EDENN SNR provides a continuous evidence score that predicts water survival under independent re-refinement substantially better than the B-factor alone. The SNR score is interpretable: $\mathrm{SNR} > 3.0$ corresponds to $\hat{m}_h > 0.90$ at the water site, equivalent to a p-value of approximately 0.001 under the null hypothesis of a noise peak.

Table 1. Comparison of ensemble-based map improvement methods. *BC conv.*: does the ensemble mean converge to a Blow–Crick map? *Unc.*: is per-voxel uncertainty a primary output? AK: averaged kick maps; ML FK: ML free-kick refinement; FEM: feature-enhanced map.

Method	Year	Perturbation	Phase distribution	BC conv.	FOM param.	Uncertainty
AK maps [20, 23]	2000/09	Coord. $\delta\mathbf{r}$	Wrapped-normal	Yes, $m(\delta)$	δ (empirical)	None
ML FK [21]	2014	Coord. $\delta\mathbf{r}$	Wrapped-normal	N/A (refinement)	$\delta \approx \hat{\sigma}_{\mathrm{coord}}$	None
FEM [22]	2015	Amplitudes w_h	Undefined	No (hist. eq.)	Many thresholds	None
EDENN	**2025**	**Atom identity**	$\mathrm{vonMises}(\varphi_{\mathrm{mod}}, \hat{X}_h)$	**Yes, optimal**	P_{omit} (data-driven)	σ_ρ, SNR

[PLACEHOLDER]: Figure 4 placeholder. ROC curves for predicting water survival under re-refinement, comparing EDENN SNR, B-factor, $2F_o - F_c$ peak height, and Phenix real-space correlation. Inset scatter: EDENN SNR vs. B-factor per water, coloured by retention outcome. AUC values with DeLong 95% CI. Expected: EDENN AUC > 0.85; EDENN outperforms B-factor by ≥ 0.07 AUC. **[Fill in from benchmark B6.]**

Mac1 active-site water network revealed by EDENN uncertainty maps

We applied EDENN to all 2,683 datasets from the Mac1 crystallographic fragment-screening campaign, encompassing the apo ground state and 214 fragment-bound structures. The EDENN SNR map of the apo state defines a reliability fingerprint of the Mac1 active site: six water molecules in the adenosine-binding region have $\mathrm{SNR} > 3.5$ across all apo datasets (constitutively present), while three waters in the catalytic-site distal ribose region show $\mathrm{SNR} \in [1.5, 3.0]$ (conditionally present, dependent on crystal batch and temperature).

[PLACEHOLDER]: Figure 5 placeholder. Panel A: Mac1 apo-state EDENN SNR map as isosurface, overlaid on protein surface. Waters coloured by SNR (< 1.5 grey, 1.5–3.0 yellow, > 3.0 blue). Panel B: Per-water SNR distribution across all apo datasets (violin plot). Panel C: Correlation of EDENN SNR with neutron-crystallography occupancy (Schuller et al. 2021) for the 12 active-site waters. Expected: Pearson $r > 0.85$. **[Fill in from Mac1 analysis, benchmark on neutron structure.]**

The EDENN difference density maps across 214 fragment-bound structures reveal a systematic pattern: fragments interacting with the adenosine site consistently produce $\mathrm{SNR}[\Delta\rho] > 3.0$ at the Asp22-coordinating water position (W1), while fragments at the catalytic distal-ribose site produce elevated SNR at W4 and W5. This clustering of water-displacement patterns by fragment chemotype defines the first experimentally quantified water pharmacophore for Mac1.

[PLACEHOLDER]: Figure 6 placeholder. Panel A: Hierarchical clustering of 214 fragment-bound datasets by their water-displacement SNR profile across 9 active-site water positions. Panel B: Representative EDENN difference density maps for two clusters, showing W1 displacement (adenosine binders) vs. W4/W5 displacement (catalytic-site binders). Panel C: Correlation of W1 displacement SNR with biochemical potency ($-\log \mathrm{IC}_{50}$, HTRF assay) for the subset of validated inhibitors. Expected: Spearman $\rho > 0.70$. **[Fill in from Mac1 EDENN analysis and ASAP assay data.]**

Single-dataset fragment detection on BAZ2B XChem data

On the 500-dataset BAZ2B fragment-screening campaign — the original PanDDA validation target with gold-standard event labels — EDENN achieves single-dataset fragment detection with recall and precision competitive with PanDDA run on the full campaign.

[PLACEHOLDER]: Figure 7 placeholder. Panel A: ROC curve of EDENN evidence ratio $\mathrm{ER}(s, d)$ vs. PanDDA campaign labels, stratified by occupancy quartile. Panel B: Precision–recall curve. Panel C: For the 47 borderline PanDDA events (Z-score 2.5–3.5, human-uncertain call), show EDENN evidence ratio and the subsequent human re-annotation outcome. Expected: EDENN AUC > 0.83 overall; recall > 0.72 for $q > 0.20$; borderline event recovery precision > 0.65. **[Fill in from BAZ2B EDENN analysis vs. PanDDA labels.]**

[PLACEHOLDER]: Table 2 placeholder. Head-to-head performance table: method, dataset size N, AUC, precision at recall=0.80, F1-score, stratified by occupancy. Methods: EDENN ($N = 1$), PanDDA ($N = 500$), PanDDA ($N = 30$), PanDDA ($N = 10$), PanDDA ($N = 1$). **[Fill in from BAZ2B benchmark.]**

EDENN is insensitive to the omit fraction across the theoretically expected range

A central practical limitation of AK maps is that the optimal kick size δ varies from 0.4–1.0 Å depending on model quality and resolution, must be tuned per dataset, and when unknown can only be averaged over at some cost to performance [20]. By contrast, equation (17) predicts that the EDENN concentration $\hat{X}_h$ decreases monotonically with P_{omit}, giving a correspondingly lower estimated FOM — a predictable, interpretable response rather than an optimisation problem.

[PLACEHOLDER]: Benchmark B4b placeholder — perturbation parameter sensitivity. For five benchmark structures (1UBQ, 1CRN, 4INS, 2LYZ, 3LZT) at R = 0.18–0.25, compute: (i) AK map CC vs F_{model} across $\delta \in \{0.1, 0.3, 0.5, 0.7, 0.9, 1.2\}$ Å using 100 kick maps. (ii) EDENN R_{map} vs the Blow–Crick reference across $P_{\mathrm{omit}} \in \{0.02, 0.05, 0.10, 0.15, 0.20, 0.30\}$ at $K = 20$. **Metric:** range of CC (AK) and R_{map} (EDENN) across parameter values. **Passing criterion:** EDENN range across $P_{\mathrm{omit}} \in [0.05, 0.20] < 0.003$; AK CC range across full δ range > 0.03. Expected: EDENN sensitivity $<$ 1/10 of AK sensitivity by ratio. **[Fill in from benchmark runs.]**

EDENN uncertainty maps passively identify model-biased regions: validation on 1zen

PDB entry 1zen [26] contains $> 10\%$ of residues displaced by a one-residue register shift in region 1–15, previously used as

a model-bias benchmark by Terwilliger et al. (2008), Pražnikar et al. [20], and Pražnikar & Turk [21]. Resolving the register ambiguity required, respectively: an iterative-build OMIT approach (Terwilliger), second-generation AK OMIT maps with manual omission of residues 1–15 and $\delta = 0.7$–1.0 Å (Pražnikar et al. 2009), and full re-refinement with ML FK (Pražnikar & Turk 2014). EDENN provides this information passively: the per-voxel σ_ρ uncertainty map should flag displaced residues without any manual intervention, because atoms refined into incorrect positions produce structurally inconsistent phases across ensemble iterations, increasing variance at those voxels.

[PLACEHOLDER]: Benchmark B-1zen placeholder — register-shift detection via per-voxel uncertainty. Protocol: run EDENN on the deposited 1zen model at $K = 20$, $P_{\text{omit}} = 0.10$ with no manual omission. Extract per-residue mean σ_ρ and $\text{SNR}[\Delta\rho]$ for all residues. Compare displaced residues (1–15, labelled from 1b57 [27] superposition) against correctly placed residues. Also compute ΔSNR between the correct register position (matching 1b57) and the deposited incorrect position in the EDENN $\Delta\rho$ map. **Metrics:** Per-residue σ_ρ AUC for predicting displaced residues (binary label from 1b57 comparison); Pearson r between σ_ρ and C^α RMSD to 1b57; ΔSNR between correct and incorrect register positions. **Passing criterion:** AUC > 0.80 without manual intervention; $\Delta\text{SNR} > 1.5$; Pearson $r > 0.55$. **Significance:** AK maps required manual omission of residues 1–15; ML FK required full re-refinement; EDENN requires neither. **[Fill in from 1zen EDENN analysis.]**

EDENN single-dataset uncertainty predicts inter-dataset conformational difference

Datasets 1RX2 (closed Met20 loop) and 1RX4 (occluded loop) of *E. coli* DHFR differ by 2% along the c-axis, rendering the classical isomorphous difference map uninterpretable while MatchMaps [24] recovers the expected conformational signal. This pair provides a test of whether EDENN single-dataset uncertainty maps, computed independently on each dataset, predict the inter-dataset structural differences that MatchMaps subsequently confirms.

[PLACEHOLDER]: Benchmark MM-1 placeholder — EDENN–MatchMaps pipeline on DHFR. Protocol: (i) Run EDENN on 1RX2 and 1RX4 independently at $K = 20$, $P_{\text{omit}} = 0.10$. (ii) Extract per-residue mean σ_ρ from each ensemble. (iii) Compare per-residue σ_ρ against C^α RMSD between the 1RX2 and 1RX4 structural models. (iv) Run MatchMaps on the 1RX2/1RX4 pair and compare $|\Delta\rho_{\text{MM}}|$ with EDENN SNR at the same voxels. **Metrics:** Pearson r between per-residue EDENN σ_ρ and C^α RMSD (1RX2 vs 1RX4); Spearman ρ between EDENN SNR and MatchMaps $|\Delta\rho|$ per voxel in the active-site and Met20-loop region; Mann–Whitney p comparing Met20-loop vs core EDENN σ_ρ. **Passing criterion:** Pearson $r > 0.60$; Spearman $\rho > 0.50$; Met20-loop mean $\sigma_\rho > 1.5\times$ core mean, $p < 0.01$. **Significance:** demonstrates that EDENN single-dataset uncertainty predicts where inter-dataset structural differences will be found, enabling principled prioritisation of MatchMaps comparisons and providing a confidence measure absent from both MatchMaps and standard maps. **[Fill in from EDENN analysis of 1RX2, 1RX4 + MatchMaps run.]**

Discussion

[PLACEHOLDER]: Opening paragraph — biological significance (3–4 sentences). Summarise the Mac1 water pharmacophore finding and its implication for inhibitor design. State the key quantitative result from Figure 6C (potency correlation). Connect to the open scientific question of why early Mac1 inhibitor series showed a potency cliff. **[Fill in after seeing Figure 6 results.]**

[PLACEHOLDER]: Comparison with PanDDA paragraph (3–4 sentences). Discuss the mechanistic reason EDENN can detect single-dataset events: FOM-free difference density vs. PanDDA amplitude subtraction. Quantify the advantage from Table 2. Explain why EDENN and PanDDA are complementary rather than competing: PanDDA excels at $N > 30$ with its σ_N advantage; EDENN excels at $N = 1$ and provides uncertainty quantification that PanDDA cannot. **[Fill in after BAZ2B results.]**

EDENN provides the theoretical foundation that the empirical kick-map tradition has lacked for three decades. Pražnikar et al. [20] identified that averaging coordinate-kicked maps improves density quality above the maximum-likelihood weighted map and empirically established the requirement for individual $|F_{\text{obs}}|$ scaling per kick iteration. The ML free-kick approach [21] independently arrived at the same mechanistic insight — that coordinate perturbation decorrelates model structure factors from working-set reflections — and exploited it for unbiased α/β estimation in refinement. Equation (16) now provides the theoretical explanation for both observations: coordinate kicking induces a wrapped-normal phase deviation whose characteristic function at $n = 1$ is exactly the Blow–Crick $m_h e^{i\varphi_{\text{mod}}}$ coefficient, but with $m_h^{\text{AK}} = \exp(-2\pi^2\delta^2|\mathbf{h}|^2/3)$, a FOM controlled by the user-chosen kick size δ rather than the data. EDENN replaces the coordinate kick with atom omission, inducing a von Mises phase distribution whose concentration $\hat{X}_h$ (17) is estimated directly from the ensemble without any free parameter, closing the circularity that requires kick-size tuning. ML FK and EDENN are complementary rather than competing: ML FK should be applied during refinement to produce a more accurately refined model with lower model bias; EDENN should be applied post-refinement to extract calibrated per-voxel uncertainty from the resulting maps. The combination offers lower phase bias at input (ML FK) and quantified uncertainty at output (EDENN), a pipeline no single prior method supports.

FEM [22] represents the most powerful empirical ensemble approach but makes a deliberate architectural choice that separates it categorically from the convergence-based framework. Histogram equalisation — FEM's defining step — collapses the amplitude information that EDENN uses as its primary uncertainty signal: a water molecule at $q = 0.05$ and a well-ordered glutamate sidechain at $q = 1.0$ are indistinguishable in an equalised map, while EDENN SNRs for the same pair differ

by a factor of ~ 20. FEM trades quantitative amplitude information for visual interpretability; EDENN preserves and formalises the information FEM discards. The community's acceptance of FEM as a map-quality improvement tool demonstrates that ensemble averaging is an established paradigm; the present work provides the theoretical framework that explains why it works and what the ensemble mean converges to — a question FEM explicitly set aside by adopting the empirical protocol.

At the inter-dataset level, EDENN occupies a distinct position from MatchMaps [24] and PanDDA [4]. MatchMaps requires two datasets and detects conformational change between them without isomorphism constraints; PanDDA requires $N \geq 30$ datasets and detects weak ligand binding by ground-state subtraction. EDENN requires one dataset and quantifies per-voxel density reliability within it. These three methods form a natural complementary toolkit. A principled pipeline for fragment screening would apply EDENN to each individual dataset to identify high-confidence single-dataset hits and flag uncertain borderline events; apply MatchMaps between the apo and hit datasets to confirm conformational changes without isomorphism assumptions; and apply PanDDA across the full campaign series to detect the weakest binders in aggregate. Benchmark MM-1 (DHFR) provides the first evidence that EDENN single-dataset σ_ρ predicts where MatchMaps inter-dataset differences will be found, suggesting that EDENN uncertainty maps can inform the design and interpretation of subsequent MatchMaps analyses.

[PLACEHOLDER]: Connection to existing uncertainty frameworks paragraph. Discuss the relationship to cryo-EM half-map FSC analysis (Nakane et al. 2020) — EDENN SNR is the crystallographic equivalent of local resolution. Discuss the relationship to ensemble refinement (Burnley et al. 2012) — EDENN provides the phase-uncertainty input that ensemble refinement requires but currently estimates empirically. **[Fill in; can largely be written now.]**

[PLACEHOLDER]: Limitations paragraph (2–3 sentences). State explicitly: von Mises approximation quality degrades for very small models ($N < 500$ atoms); the CLT argument underpinning Theorem 2 requires validation in this regime. State the computational cost: $K = 20$ EDENN iterations takes approximately X minutes per dataset on a standard workstation. Note that EDENN does not address the problem of poor data quality (low completeness, radiation damage) independently of model quality. **[Fill in after benchmarking compute time.]**

[PLACEHOLDER]: Future directions paragraph. Note the natural extension to cryo-EM phase uncertainty (replacing model phases with half-map derived phases). Note the potential to supervise AlphaFold-initialized model building (ModelAngelo, Terwilliger et al. 2023) with EDENN-derived phase distributions. Note the planned integration of ML FK refinement upstream of EDENN analysis as a best-practice pipeline. Note the planned integration with ASAP's continuous design-make-test pipeline. **[Fill in; can largely be written now.]**

Methods

EDENN implementation

[PLACEHOLDER]: Software paragraph. Describe the Python implementation: gemmi for MTZ I/O and structure factor computation, numpy for ensemble statistics, CCP4-compatible output. State the GitHub repository URL and version used for all analyses. Conda install one-liner. **[Fill in once software is finalised.]**

Omit fraction selection

The omit fraction P_{omit} controls the trade-off between phase perturbation magnitude and convergence rate. From equation (9), the concentration of the induced von Mises distribution is:

$$X_h^{\text{EDENN}} \approx \frac{2(1-P_{\text{omit}})}{P_{\text{omit}}} \cdot \frac{|F_{\text{mod}}(\mathbf{h})|^2}{\sum_j |f_j(s)|^2} \tag{17}$$

For $P_{\text{omit}} = 0.10$ and a typical protein at 2 Å resolution, this yields $X_h \approx 2$–5, giving $m_h \approx 0.60$–0.85, consistent with observed crystallographic FOM ranges. We recommend $P_{\text{omit}} \in [0.05, 0.15]$ and $K = 20$ as the default, sufficient for $R_{\text{map}} < 0.01$ convergence.

Convergence assessment

Convergence is monitored by computing the running R_{map} between $\langle\rho\rangle_{\text{omit}}^{(k)}$ and $\langle\rho\rangle_{\text{omit}}^{(K)}$ as k increases. We fit $R_{\text{map}}(k) = a/\sqrt{k} + b$ and require $R^2 > 0.95$ and asymptotic bias $b < 0.01$ as passing criteria.

Datasets and data processing

NSP3 Mac1 fragment screening campaign

[PLACEHOLDER]: Mac1 data paragraph. State the source of the MTZ files (ASAP internal repository). Dataset statistics: number of soaking experiments, resolution range, space group, unit cell parameters. PanDDA version and parameters used for ground-state estimation. **[Fill in with actual ASAP data provenance.]**

BAZ2B bromodomain campaign

[PLACEHOLDER]: BAZ2B data paragraph. Dataset provenance (XChem/Diamond/SGC). Number of datasets, resolution range. PanDDA ground-truth labels: how events were called, human annotation protocol, borderline event definition. **[Fill in with BAZ2B data provenance.]**

Benchmark structures

Five high-resolution benchmark structures were downloaded from the PDB (accessions 1UBQ, 1CRN, 4INS, 2LYZ, 3LZT) and their structure factors retrieved. REFMAC5 [12] was run with default parameters to generate Sigma-A FOM-weighted maps for comparison. SAD phases from 5FIO were used as experimental ground truth for FOM calibration benchmarks.

Statistical analysis

Map distance metrics

The Fourier-space L^2 distance (R-map) between maps ρ^A and ρ^B is:

$$R_{\text{map}} = \frac{\sum_h |w_h^A F_h^A - w_h^B F_h^B|}{\sum_h |w_h^A F_h^A + w_h^B F_h^B|} \tag{18}$$

Fourier shell correlation is computed per resolution shell using equation (4) from ref. [13]. Real-space Pearson correlation (RSCC) is computed over all voxels within the asymmetric unit.

Water survival analysis

Waters from deposited models were classified as "retained" if a water appeared within 0.5 Å of any automatically placed water in an independent re-refinement with Phenix [14]. AUC comparison between EDENN SNR and B-factor used the DeLong method with bootstrap confidence intervals ($n = 10{,}000$ resamples) [17].

Fragment event analysis

The evidence ratio for fragment binding is defined as:

$$\text{ER}(s,d) = \frac{\text{SNR}[\Delta\rho_d](s)}{\text{SNR}[\Delta\rho_{\text{null}}](s)} \tag{19}$$

where the null distribution of $\text{SNR}[\Delta\rho]$ is estimated from outer-shell reflections ($d < 4.0$ Å) where no specific fragment signal is expected. Events were called at $\text{ER} > 3.0$, corresponding to a false positive rate < 0.05 under the null model (calibrated empirically on known apo datasets). McNemar's test was used for paired binary outcome comparison between EDENN and PanDDA [18].

Power analysis

Statistical power for detecting a fragment at occupancy q with K ensemble iterations:

$$\text{Power}(q,K) = \Phi\left(\frac{qS/\sigma_{\text{ens}}(K) - z_\alpha}{\sqrt{1/K}}\right) \tag{20}$$

where S is the mean fragment structure factor amplitude, $\sigma_{\text{ens}}(K)$ is the ensemble noise estimated empirically, $z_\alpha = 1.645$ for $\alpha = 0.05$, and Φ is the standard normal CDF.

Benchmarking protocol

Seven formal benchmarks were defined prior to data analysis (pre-registration). Benchmarks B1–B4 establish mathematical properties of the method (convergence, FOM calibration, phase distribution calibration, convergence rate). Benchmarks B5–B7 establish biological utility (fragment detection, water scoring, conformational uncertainty). All passing criteria were set before data collection: $R_{\text{map}} < 0.01$ at $K = 20$ (B1); per-shell FOM bias < 0.05 (B2); coverage within $\pm 5\%$ of nominal (B3); water AUC > 0.80 (B5); fragment AUC > 0.75 on BAZ2B (B6).

Data availability

[PLACEHOLDER]: All Mac1 MTZ files used in this study are available through the ASAP open-science data portal at `https://asapdiscovery.org`. BAZ2B structure factors are available through the PDB (accessions listed in Supplementary Table S1). Processed EDENN maps for all benchmark and application datasets are deposited at [ZENODO DOI, to be assigned]. **[Fill in DOIs at submission.]**

Code availability

[PLACEHOLDER]: EDENN is available as an open-source Python package at `https://github.com/[organisation]/edenn` (v[X.Y.Z] used here). Install with `conda install -c conda-forge edenn`. A worked example reproducing Figure 1 on PDB entry 1UBQ is provided in the repository. **[Fill in repository URL and version at submission.]**

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Author contributions

[PLACEHOLDER]: H.K.: mathematical framework, software implementation, all computational analyses. C.W.: PanDDA 2.0 integration, BAZ2B ground-truth labels, XChem data pipeline expertise. F.v.D.: scientific direction, ASAP data access, Mac1 campaign design, supervision. **[Revise according to actual contributions.]**

Competing interests

The authors declare no competing interests.

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Figure Legends

Figure 1 | EDENN ensemble mean converges to the Blow–Crick optimal map. a, R_{map} between the EDENN ensemble mean and the REFMAC FOM-weighted map as a function of ensemble size K, shown for five benchmark structures. Points are means across all reflections; error bars are s.d. across 5 independent runs with different random seeds. Dashed line shows $K^{-1/2}$ scaling. **b,** Per-reflection scatter plot of EDENN FOM $\hat{m}_h$ vs. REFMAC Sigma-A FOM for the same structures, coloured by resolution shell. **c,** Fourier shell correlation profiles at $K = 5$ (dashed) and $K = 50$ (solid), averaged across all benchmark structures.

[PLACEHOLDER]: [Replace with actual figure.]

Figure 2 | EDENN FOM estimation is unbiased relative to SAD experimental ground truth. Resolution-shell FOM profiles for EDENN, REFMAC Sigma-A, Lunin LB, and empirical estimates from SAD phases, for three test structures. Shaded bands show 95% bootstrap confidence intervals.

[PLACEHOLDER]: [Replace with actual figure.]

Figure 3 | EDENN phase distributions are calibrated to nominal coverage. Calibration plot of empirical coverage vs. nominal level for von Mises credible intervals at 68%, 90%, and 95%. Points are per-resolution-shell means; dashed line is the perfect-calibration diagonal.

[PLACEHOLDER]: [Replace with actual figure.]

Figure 4 | EDENN SNR predicts genuine crystallographic waters better than B-factor. a, ROC curves for predicting water survival under independent re-refinement, for EDENN SNR, B-factor, $2F_o - F_c$ peak height, and Phenix RSCC. AUC values (DeLong 95% CI) are indicated. **b,** EDENN SNR vs. B-factor per water, coloured by survival outcome.

[PLACEHOLDER]: [Replace with actual figure.]

Figure 5 | EDENN uncertainty maps of the Mac1 active-site water network. a, EDENN SNR isosurface map of the Mac1 apo active site, with waters coloured by SNR category. **b,** Per-water SNR distribution across all apo datasets. **c,** Correlation of EDENN SNR with neutron-crystallography occupancy for 12 active-site waters.

[PLACEHOLDER]: [Replace with actual figure.]

Figure 6 | Fragment-induced water displacement patterns define a Mac1 pharmacophore. a, Hierarchical clustering of 214 fragment-bound datasets by their water-displacement SNR profile. **b,** Representative EDENN difference density maps for two chemotype clusters. **c,** Correlation of W1 displacement SNR with biochemical potency for validated inhibitors.

[PLACEHOLDER]: [Replace with actual figure.]

Figure 7 | Single-dataset fragment detection on BAZ2B bromodomain. a, ROC curve of EDENN evidence ratio vs. PanDDA gold-standard labels, stratified by occupancy. **b,** Precision–recall curve. **c,** EDENN evidence ratios for the 47 borderline PanDDA events, annotated by final human classification.

[PLACEHOLDER]: [Replace with actual figure.]

Supplementary Information

Supplementary Note 1: Proof of Lemma 3 (CLT for omit phases)

For atoms $\{Z_j^{(k)}\} = \{\mathbf{1}[j \notin \mathcal{O}_k] f_j(s) e^{2\pi i \mathbf{h}\cdot\mathbf{r}_j}\}$, each $Z_j^{(k)}$ is independent across j with $|Z_j^{(k)}| \le f_j^{\max}$ bounded. The Lyapunov CLT applies, giving:

$$F_{\mathrm{omit}}^{(k)}(\mathbf{h}) - (1 - P_{\mathrm{omit}}) F_{\mathrm{mod}}(\mathbf{h}) \xrightarrow{d} \mathcal{N}_{\mathbb{C}}(0, \Sigma_h)$$

where $\Sigma_h = P_{\mathrm{omit}}(1 - P_{\mathrm{omit}}) \sum_j |f_j(s)|^2$. For a complex Gaussian perturbation of amplitude A and phase μ with variance σ^2, the induced phase distribution is von Mises$(\mu, A^2/\sigma^2)$ to leading order in σ/A [**?**]. Substituting gives (17). □

Supplementary Note 2: Centric reflections

For centric reflections ($\varphi_h \in \{0, \pi\}$), the von Mises distribution is replaced by a Bernoulli distribution and I_1/I_0 is replaced by $\tanh(X_h/2)$. All theorems above hold with this substitution.

Supplementary Table S1

[PLACEHOLDER]: Full list of PDB accessions and ASAP dataset identifiers used in each analysis. **[Fill in at submission.]**

Supplementary Table S2

[PLACEHOLDER]: Complete benchmark results table including all B1–B7 metrics, per structure, with confidence intervals. **[Fill in after benchmark runs.]**